QUESTIONS

1. Explain each of these terms:
   1 Inheritance
   2 Genetics
   3 Chromosome
   4 DNA
   5 Characteristics
   6 Meiosis
   7 Gamete
   8 Fertilisation
   9 Zygote
   10 Homologous chromosomes
   11 Gene
   12 Allele
   13 Locus
   14 Pure-breeding
   15 Hybrid
   16 Punnett square
   17 Incomplete dominance
   18 Sex determination
   19 Sex chromosome
   20 *Gregor Mendel
   21 *Hypothesis
   22 *Breeding experiment
   23 *Test cross
   24 *Law of Segregation
   25 *Dihybrid cross
   26 *Law of Independent Assortment
   27 *Linkage
   28 *Sex-linked genes
   29 *X-linked traits
   30 *Colour blindness
   31 *Haemophilia
   32 *Carrier female
   33 *Non-nuclear inheritance

2. Distinguish between the following pairs of terms:
   (a) Mitosis and meiosis
   (b) Dominant and recessive alleles
   (c) Genotype and phenotype
   (d) Haploid and diploid
   (e) Self-pollination and self-fertilisation
   (f) Homozygous and heterozygous
   (g) F_1 and F_2 generation
   (h) Anther and stigma
   (i) Homozygous dominant and homozygous recessive
   (j) XX and XY
   (k) Homologous and homozygous
   (l) Cross-pollination and self-pollination

3. Using the symbols N = normal wing and n = short wing, write down the following:
   (a) The genotype of a fly which is heterozygous for this trait.
   (b) The possible genotypes of its gametes.
   (c) Work out what kind of offspring would be produced if a heterozygous fly mated with one which was homozygous for normal wing.

4. A red-flowered tulip was crossed with a white-flowered tulip. The seeds were collected and all grew into red tulips.
   (a) Which colour was dominant?
   (b) What was the genotype of the offspring?
   (c) Some of the offspring were self-fertilised. The plants produced had three times as many red as white flowers. Use a diagram to explain this cross.

5. In humans, brown eyes is usually dominant over blue eyes. A blue-eyed man marries a brown-eyed woman whose father was blue-eyed. What percentage of their children should have blue eyes?

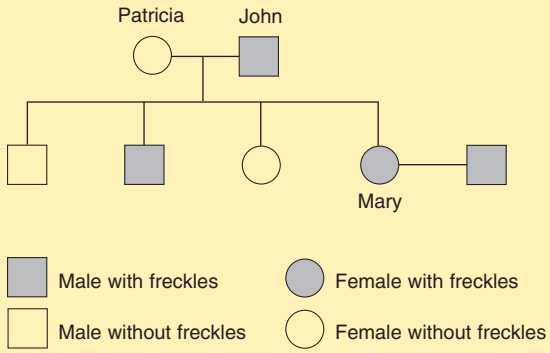
6. In a certain type of plant, white fruit colour is dominant over yellow. A fruit-grower crosses a white-fruited plant with a yellow-fruited one. About half the offspring have white fruit and half have yellow fruit.
   (a) What are the genotypes of the parent plants? Use diagrams to show your reasoning.
   (b) If you were to self-pollinate one of the white-fruited offspring, what phenotypes would you expect to obtain in the offspring and in what proportions?

7. When two white cats were mated they produced black kittens as well as white kittens. Show using diagrams how this can arise.

8. A brown-eyed man, whose father was brown-eyed and mother was blue-eyed, marries a blue-eyed woman whose father and mother were both brown-eyed. The couple has a blue-eyed son. Give the genotypes of each of the five people.

9. Mr and Mrs Lane have three children, all boys. They are sure that their next child will be a girl. Do you agree? Explain your answer.

10. Freckles on the skin is an inherited characteristic in humans. The allele for freckles (F) is dominant over the allele for absence of freckles (f). The diagram represents part of a family tree showing the inheritance of freckles.

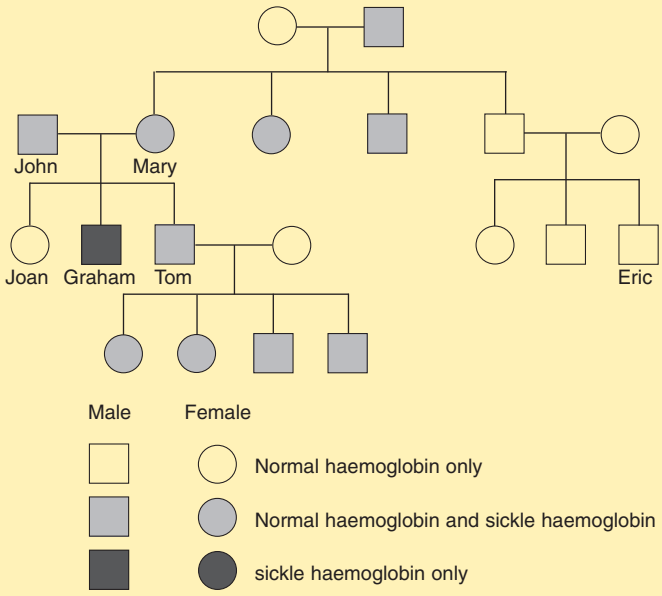


(a) Give the possible genotypes of John and Patricia.
(b) Mary marries a man homozygous for freckled skin. Show the possible phenotypes and genotypes of their children.

11. In a study of garden peas, red-flowered plants were crossed with white-flowered plants. The first generation of plants all had red flowers. However, when these red-flowered plants were allowed to self-fertilise, about 25% of the offspring had white flowers, the remainder having red flowers. In a similar investigation with snapdragon plants, when red-flowered plants were crossed with white flowered plants, the resulting first generation all had pink flowers. When these pink-flowered plants were self-fertilised, 25% of the offspring had white flowers, 25% had red flowers and 50% had pink flowers.
Using diagrams, show why the results obtained with the garden pea are different from those obtained with the snapdragon plants.

12. In guinea pigs the characteristic white coat colour (b) is recessive to black coat colour (B).
(a) A pure-breeding black guinea pig is crossed with a pure-breeding white guinea pig. Show the genotypes of the gametes, and the genotypes and phenotypes of the progeny of the cross.
(b) You are given a black male guinea pig of unknown genotype and asked to find out whether it is homozygous or heterozygous. What crosses would you carry out? Show your reasons by using a diagram.

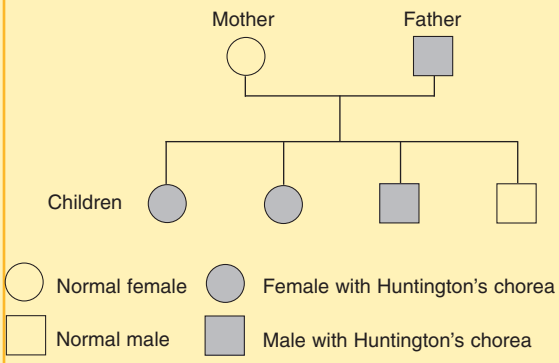
13. The gene that controls the formation of haemoglobin exists in two forms. The normal form of the gene (H) produces normal haemoglobin. The mutant form of the gene (h) produces 'sickle' haemoglobin. People who are homozygous for the mutant form of the gene suffer from a condition called sickle cell anaemia. When their haemoglobin gives up its oxygen, the red blood cells change to a sickle shape. These sickle-shaped cells may stick together and block small blood vessels, often causing death. People who are heterozygous for the gene produce both normal and 'sickle' types of haemoglobin, but do not usually suffer from sickle cell anaemia. The family tree shows the inheritance of haemoglobin type.



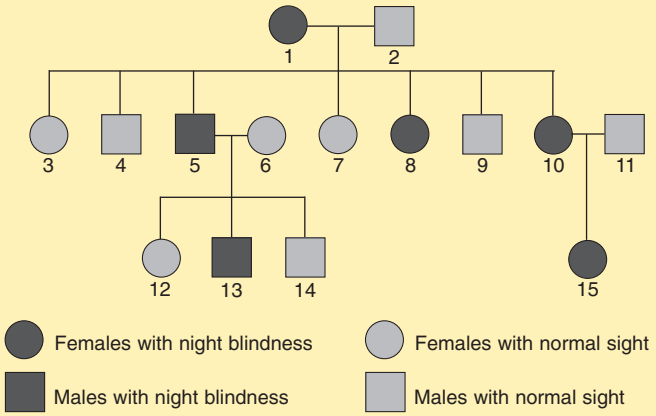
(a) Copy the chart and write inside each square or circle the possible genotypes of each person.
(b) Cousins Joan and Eric wish to marry but are worried their children may inherit sickle cell haemoglobin. What advice would you give to them? Explain the reason for your answer.

14. When true-breeding red-flowered plants are crossed with true-breeding white ones of the same species the offspring (F_1 generation) all have red flowers. Show using diagrams, what types of flower would you expect to be produced, if these F_1 plants are:
(a) self-pollinated and the resulting seeds sown.
(b) cross-pollinated with white flowers and the resulting seeds sown.
(c) propagated vegetatively?
Give reasons for your answers (show the crosses).

15. Huntington's chorea is a severe condition, which causes mental deterioration and loss of muscular coordination. It is caused by the inheritance of a dominant allele. The effects of this allele do not normally appear until a person is at least over 30 years old. The diagram shows a family pedigree for the inheritance of Huntington's chorea.
    (a) Copy the chart and write inside each square or circle the possible genotypes of each person.
    (b) Show how it is possible for 'normal' offspring to be born even though one parent was affected. (Use H for the allele for Huntington's chorea and h for the recessive allele.)
    (c) It is possible for a HH person to be unaware of the disease and pass it on to their children. Explain how this can happen.

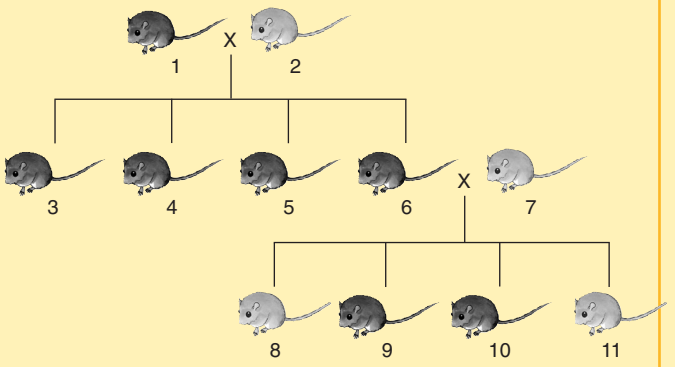


16. Look at the family tree. Use N as the symbol for the night-blindness allele (dominant), and n for the normal allele (recessive).



    (a) Copy the chart and write inside each square or circle the possible genotypes of each person.
    (b) If persons 13 and 15 should marry, what is the chance that any of their children will be night-blind? Explain your answer.
    (c) If persons 14 and 15 should marry, what is the chance that any of their children will be night-blind? Explain your answer.

17. The diagram represents an experiment involving black and white mice. The parents 1 and 2 had many more offspring than are shown, and all of them were black.



    (a) If the genotype of animal number 5 is Bb, state the genotypes of animals number 1, number 7 and number 9.
    (b) In this experiment, what is the phenotype of an animal which is homozygous recessive?
    (c) List the numbers of all the animals shown in the diagram which are homozygous.
    (d) Using diagrams, show the phenotypes of the offspring produced as a result of crossing animals 9 and 10.

18. Find out more about Gregor Mendel, then write a short letter to him bringing him up to date on present day genetics.

19. Using identical twins, how might you test the hypothesis that intelligence is inherited from parents rather than acquired from the environment as they grow up.

20. Explain the basis of sex determination in humans.

21. In humans an individual's sex is determined by genes located on pairs of chromosomes called sex chromosomes. The male pair is shown as XY and the female pair as XX. Show that there is an equal chance of a child being male or female by copying and completing the Punnett square in your answer book. Give the gametes of each parent and the genotypes of the offspring.

(LC OL Sample Paper)

22. (a) Explain the following terms that are used in genetics – dominance, genotype, phenotype.
    (b) In Aberdeen Angus cattle, the polled (P) condition (absence of horns) is dominant to the horned (p) condition. A heterzygous polled bull was crossed with a horned cow. Use the following layout in your answer book to find the possible genotypes and phenotypes of the calves that may result from this cross.

	Heterozygous polled bull	x	Horned cow
Genotype of parents	_____		_____
Gametes	_____		_____
Genotypes of calves	_____		_____
Phenotypes of calves	_____		_____

(*LC OL Exam Paper*)

HIGHER LEVEL

23. A woman has a haemophilic son and three normal sons.
    (a) What is her genotype and that of her husband with respect to this gene? Explain your answer using diagrams.
    (b) Could she have a haemophilic daughter? Explain your answer.

24. A population of humans will contain many more colour-blind individuals than haemophiliacs although the genes are transmitted in the same way. Explain this difference in frequency.

25. In guinea pigs black coat colour (B) is dominant to white coat colour (b), and short hair (S) is dominant to long hair (s). A black-coated short-haired guinea pig, heterozygous for both characteristics, is mated with a white-coated long-haired guinea pig. Show, using diagrams, the genotype and phenotype of the possible offspring.

26. In snapdragons the flower colour can be red (R), white (r) or, in the heterozygous condition, pink. In addition, tall (T) is dominant over dwarf (t). A dwarf red snapdragon plant is crossed with a homozygous tall white snapdragon plant. Show, using diagrams, the genotype and phenotype of the possible offspring.

27. A sex-linked recessive gene produces red-green colour blindness in humans. A normal woman, whose father was colour blind, marries a colour-blind man.
    (a) Show the possible genotypes for the mother of the colour-blind man.
    (b) What are the chances that their first child will be a colour-blind boy?
    (c) Of all the girls produced by these parents, what % are expected to be colour-blind?
    (d) What % of their children are expected to be normal?

28. Sweet pea plants with purple flowers and long pollen grains were crossed with plants with red flowers and round pollen grains. The F_1 plants were all purple-flowered with long pollen grains. When these F_1 plants were back-crossed with red flowered, round-pollen plants, the resulting offspring were:

2,334	purple long
23	purple round
24	red long
2,296	red round

    This is a ratio of about 20 : 1 : 1 : 20
    (a) State Mendel's 2nd Law of Independent Assortment.
    (b) According to this law, what ratio would have been expected in the F2 generation?
    (c) Suggest why the results were not as expected.

29. In humans, haemophilia depends on the recessive allele of a sex-linked gene (80,000 cases in the US).
    (a) A man, whose father was a haemophiliac, but whose own blood clotting time is normal, marries a normal woman with no record of haemophilia in her ancestry. What is the chance of haemophilia among their children?
    (b) A woman, whose father was a haemophiliac, but who is not herself a 'bleeder', marries a normal man. What is the chance of haemophilia among their children?

30. (a) (i) What is meant by the term 'segregation' as used in genetics?
(ii) State the products of 'segregation' when an organism is heterozygous in respect of the dominant allele A.

(b) (i) State the Law of Independent Assortment.
(ii) In cattle the allele for red coat (R) is dominant to the allele for black coat (r) and the allele for straight coat (S) is dominant to the allele for curly coat (s). When a bull with a red, straight coat was mated with cows with curly, black coats it was found that calves with four different phenotypes resulted. These four phenotypes occurred in equal numbers.
1. State the genotypes of the bull and of the cows.
2. State the phenotypes of the calves.
3. Which phenotypes of the calves suggest that independent assortment has taken place?
4. Would you expect different phenotypes if the genes for coat colour and coat type were located on the same chromosome? Explain your answer.

(LC HL Sample Paper)

31. In tomato plants the allele responsible for purple stem (P) is dominant to that for green stem (p) and the allele for cut leaf (C) is dominant to the allele for potato-type leaf (c). A plant with a purple stem and cut leaves was crossed with a plant with a green stem and potato-type leaves. A total of 448 seeds was obtained. When the seeds were germinated four types of progeny resulted and they had the following phenotypes:

110	purple stem and cut leaves
115	green stem and potato-type leaves
114	purple stem and potato-type leaves
109	green stem and cut leaves

(a) What were the genotypes of the tomato plants that gave rise to these progeny?

(b) Do the progeny of this cross illustrate the Law of Independent Assortment? Explain your answer.

(LC HL Exam Paper)